DIAGRAM ATLAS · PLATES 01–13

ZeroGate

A zero-trust access broker for web applications and SSH servers. Thirteen diagrams covering the architecture, domain, schema, flows, decision logic, and state machines.

FRAMEWORK
NIST SP 800-207

SCOPE
Web apps · SSH

STATUS
Design baseline v0.9

PLATES
13 + cover

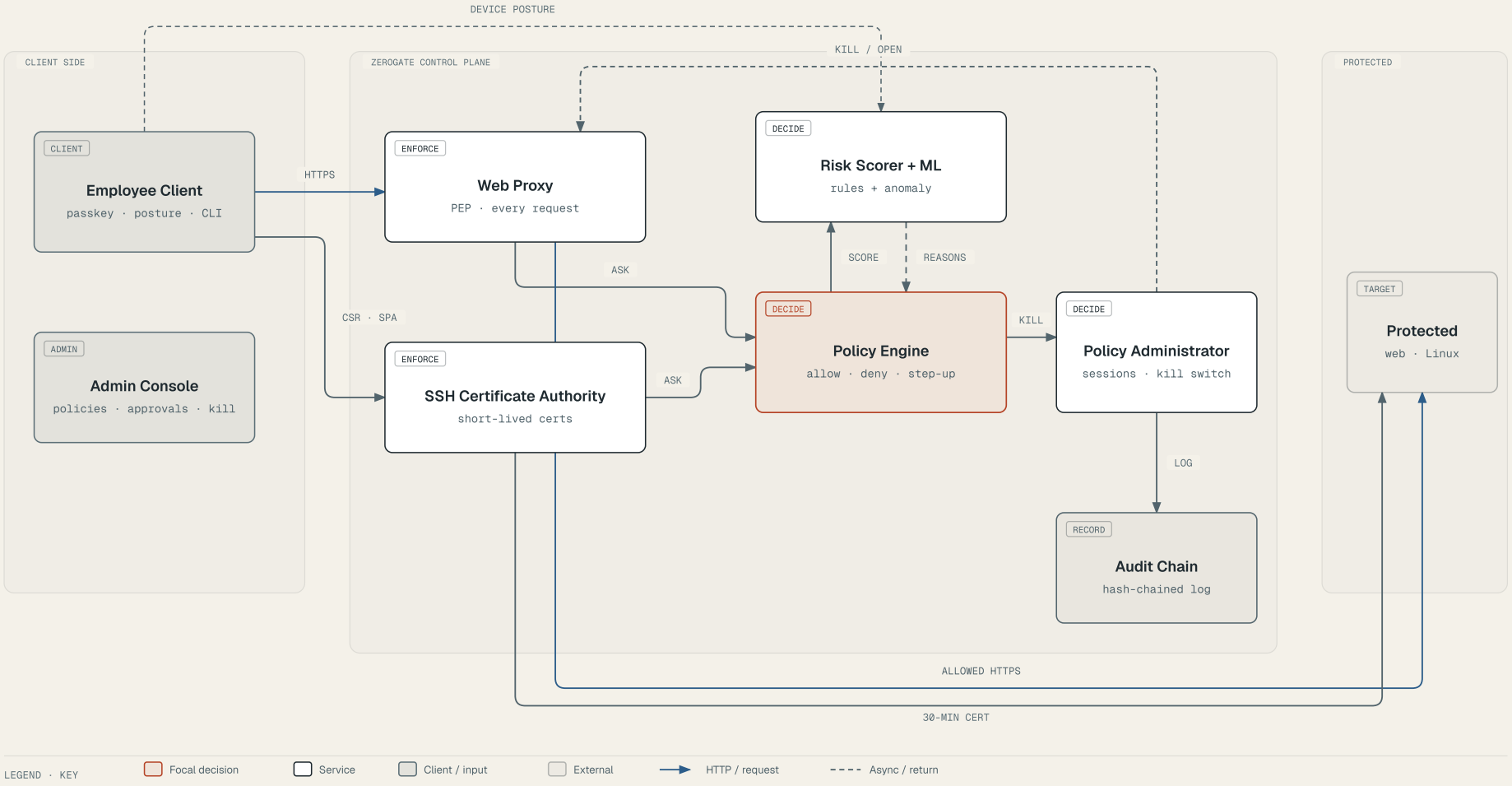
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Vector plates. Every figure is also available as a separate PDF at its native page size.

Component architecture

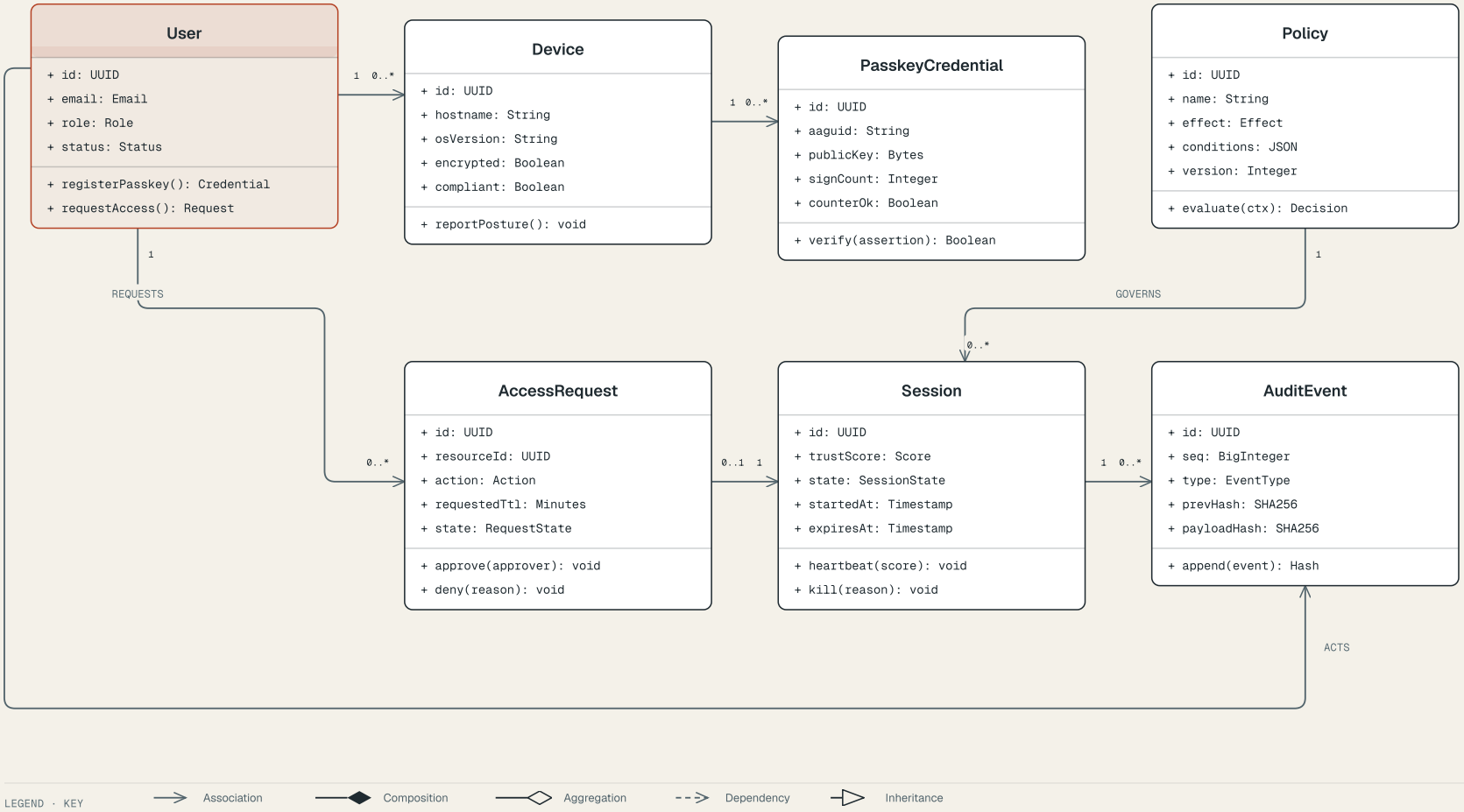
COMPONENT VIEW



Five zones, two enforcement points, one policy decision. Requests enter through the Web Proxy or the SSH Certificate Authority and never reach a resource without a decision.

Domain model

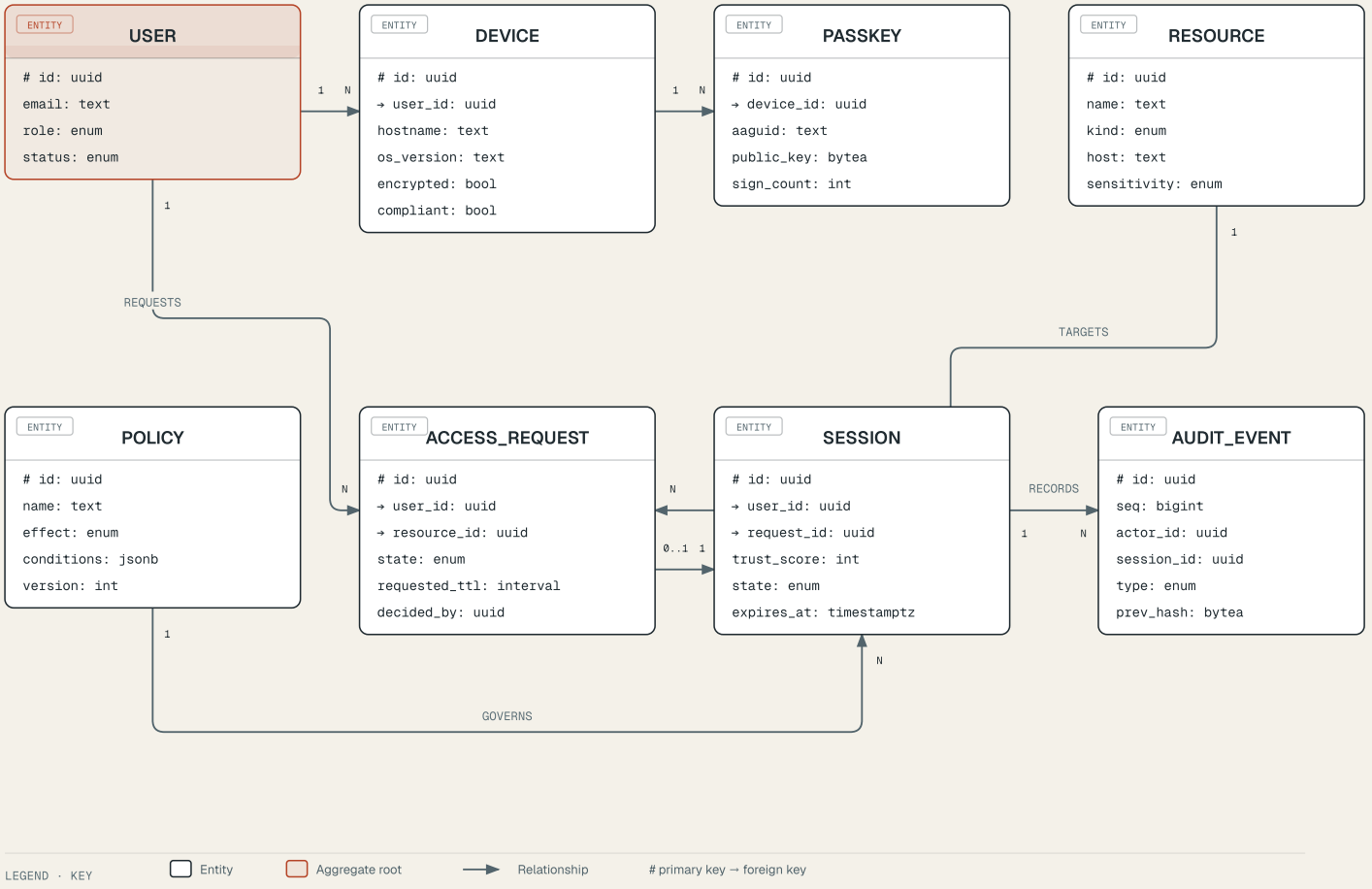
UML CLASS DIAGRAM



The seven domain objects and the invariants they protect: identity, device, credential, resource, policy, request, session, and the audit record that ties them together.

Logical data model

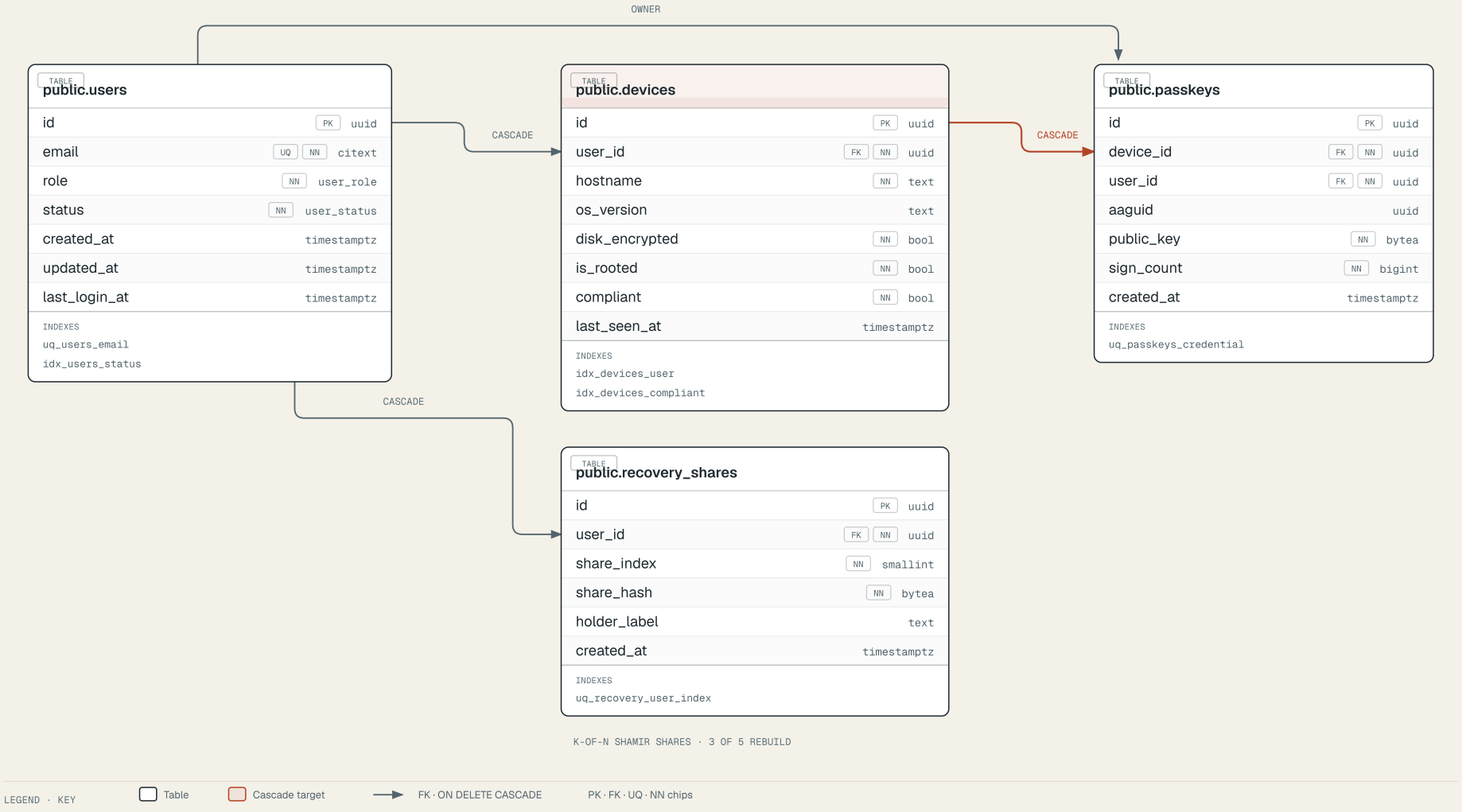
ENTITY-RELATIONSHIP



Entities, keys, and cardinalities across identity, policy, and access. The two physical schemas that follow resolve this model into PostgreSQL tables.

Identity and device schema

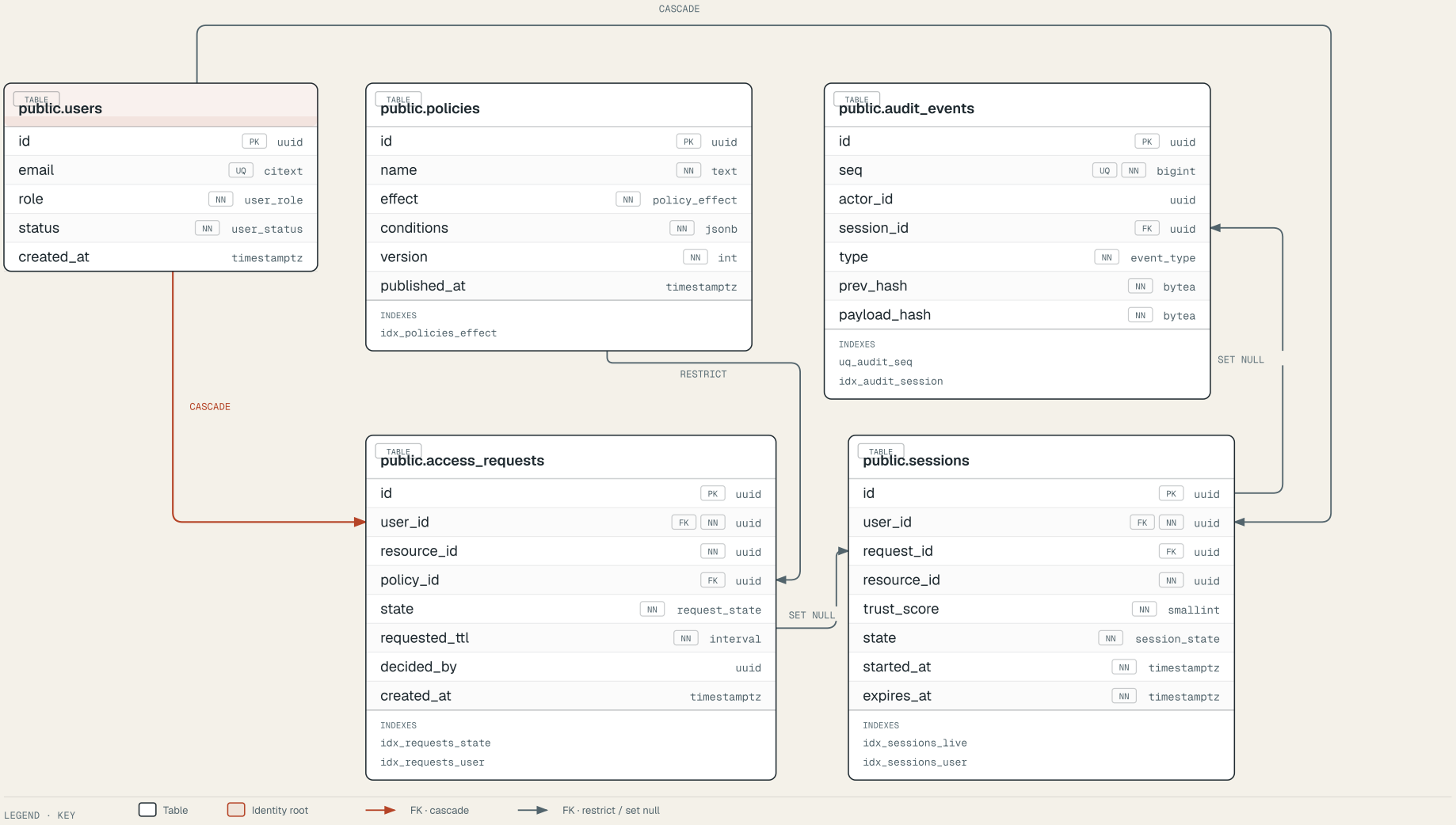
PHYSICAL SCHEMA



Users, devices, passkeys, and recovery shares. Every foreign key is explicit and every delete behaviour is stated on the edge.

Access, policy and audit schema

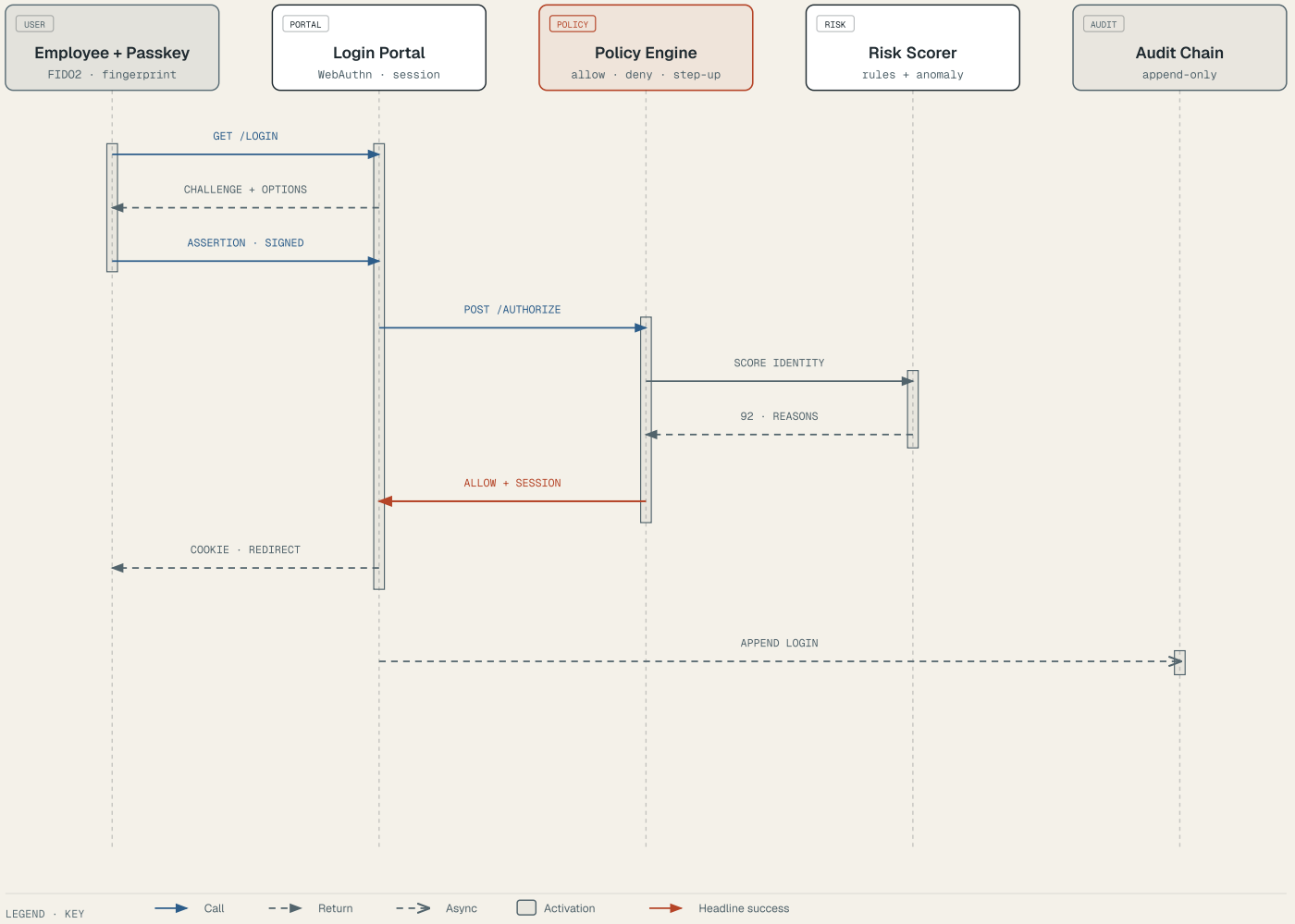
PHYSICAL SCHEMA



Policies, access requests, sessions, and the append-only audit chain. Five foreign keys carry the cascade, restrict, and set-null rules.

Passkey login sequence

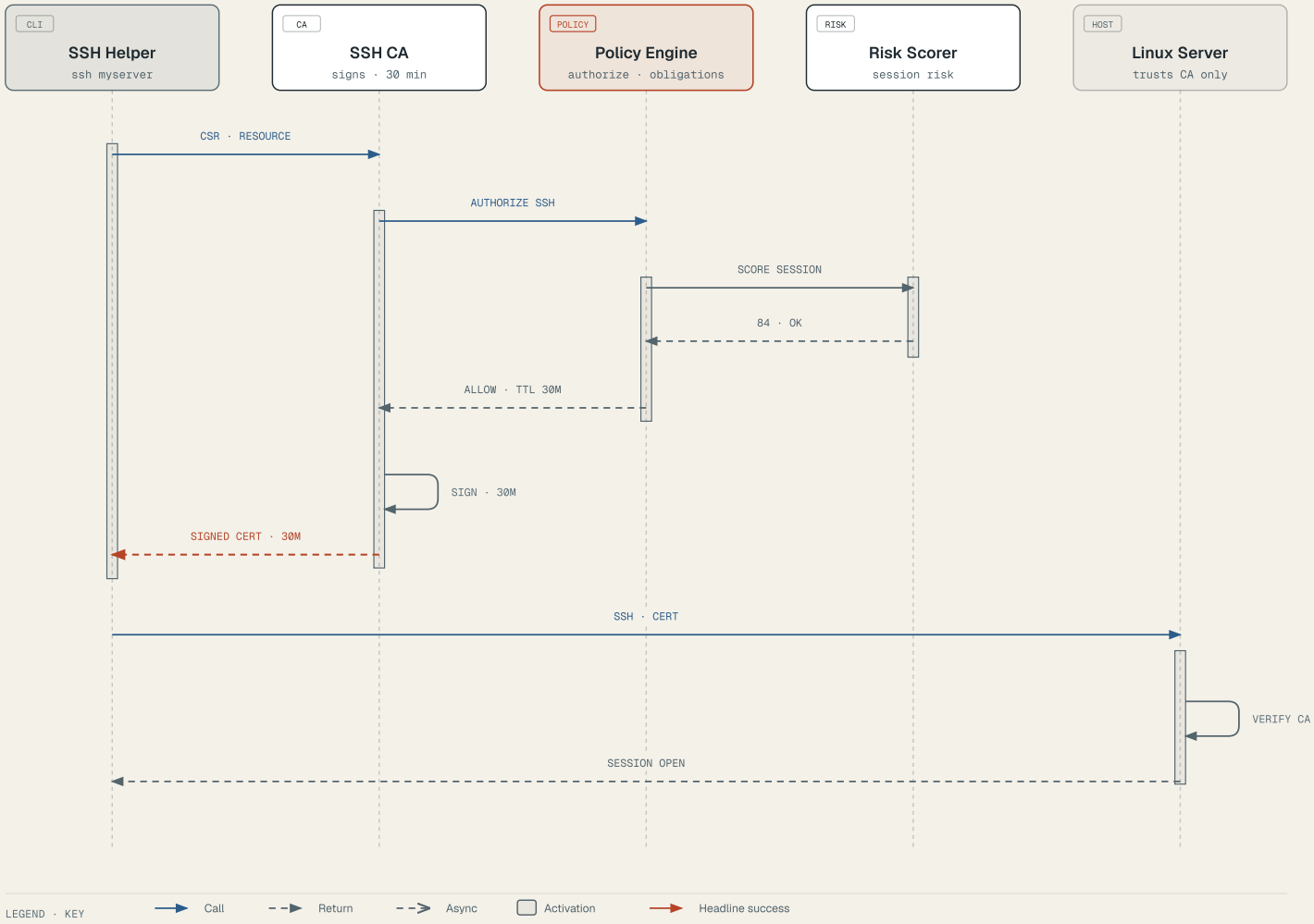
SEQUENCE DIAGRAM



A WebAuthn ceremony, a risk decision, and a server-side session cookie. No password is transmitted and no biometric leaves the device.

SSH certificate issuance

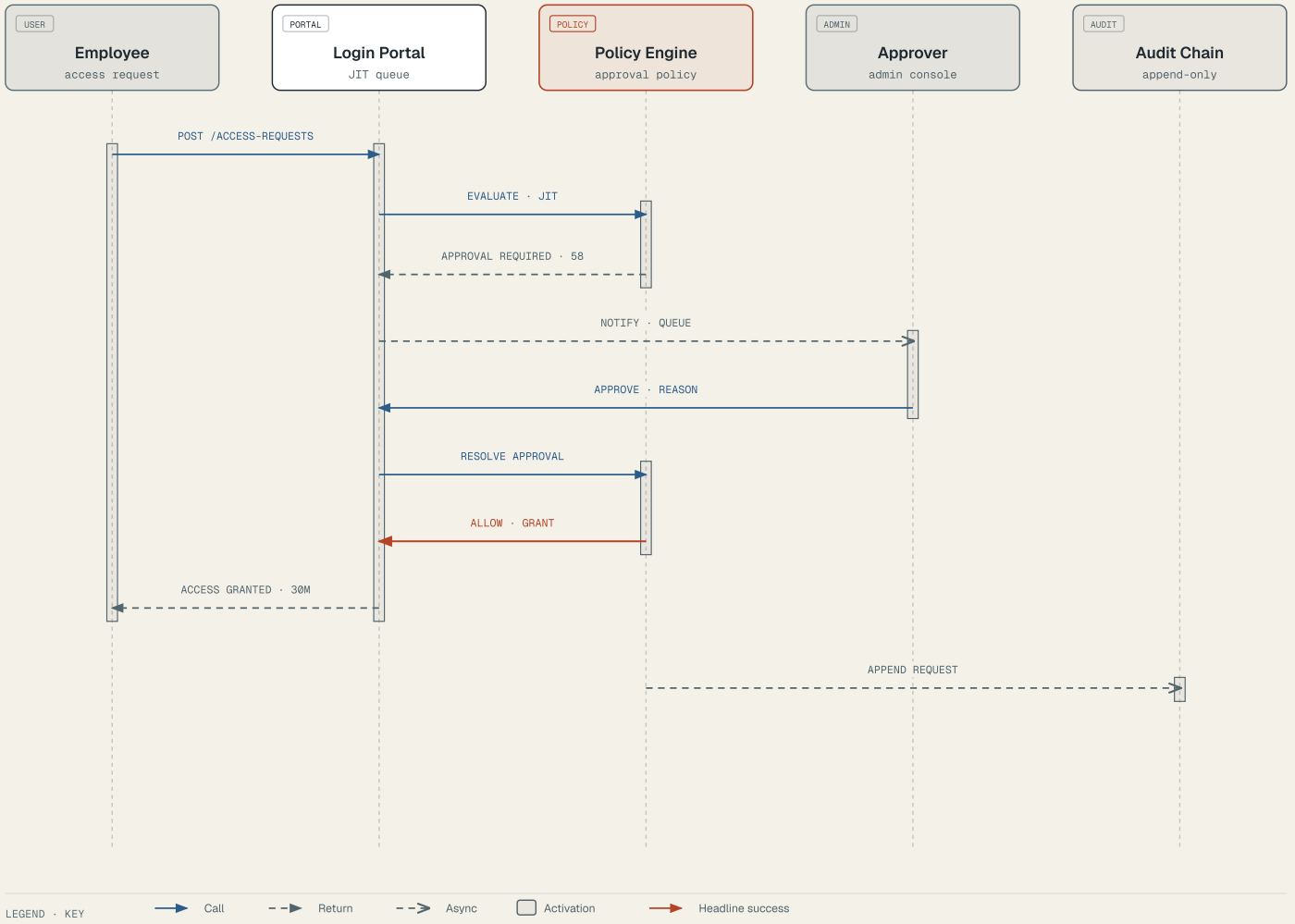
SEQUENCE DIAGRAM



The SSH helper proves possession, requests a thirty-minute certificate for one principal and one host, and opens the SPA port only long enough to connect.

Just-in-time approval

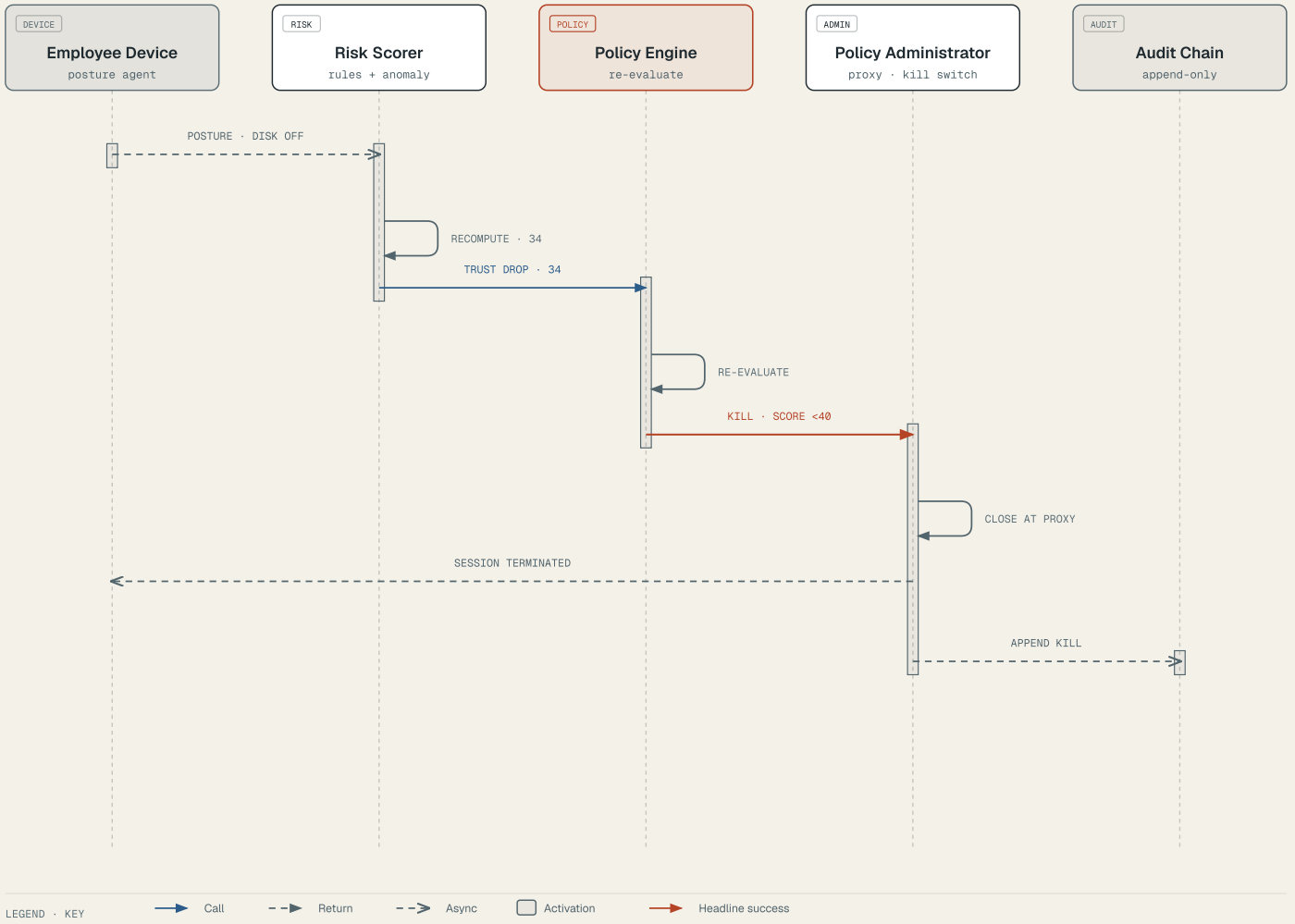
SEQUENCE DIAGRAM



A step-up score becomes a human decision. The approver records a reason; the grant is capped at thirty minutes and the queue waits fifteen before it expires.

Mid-session kill

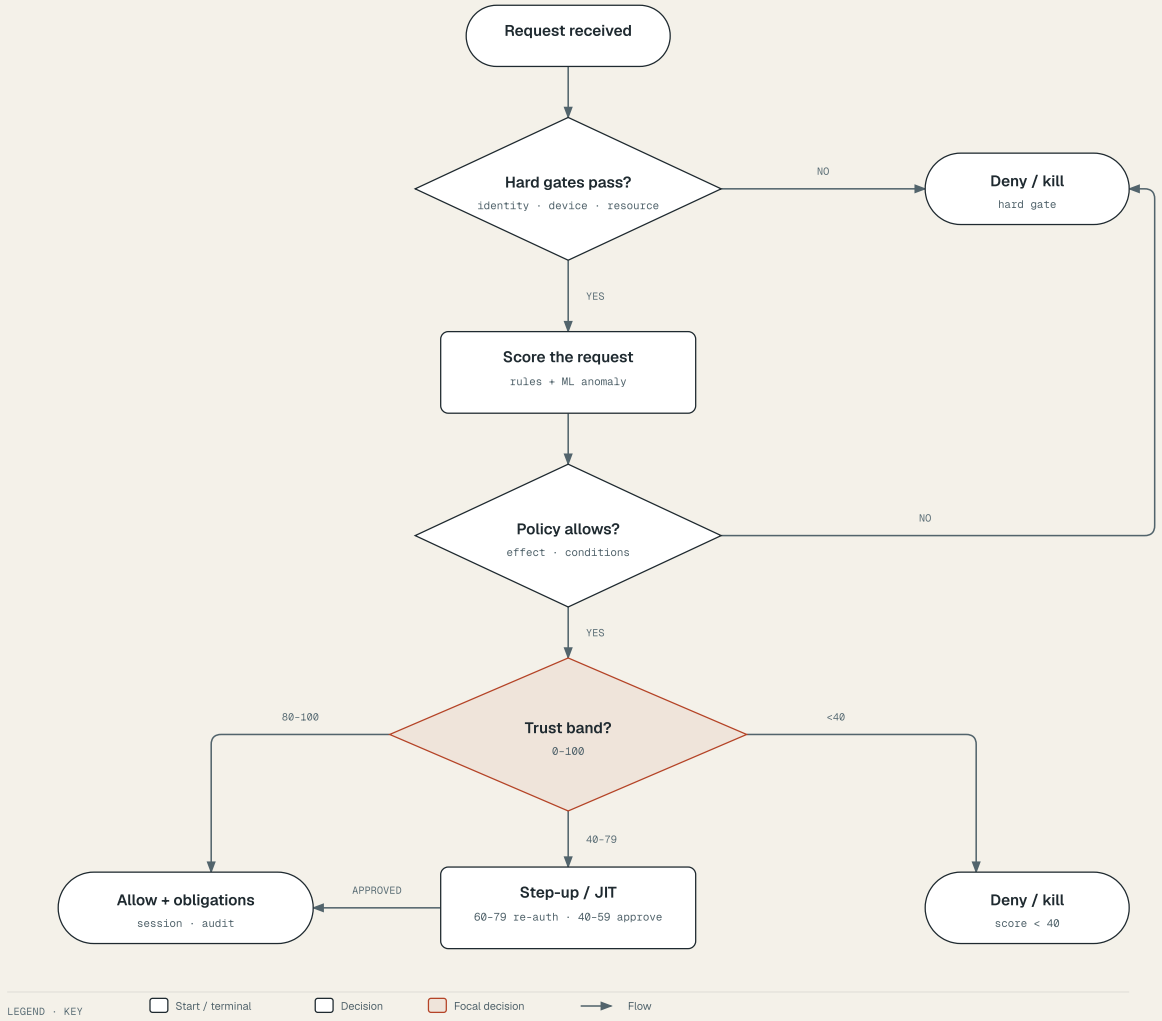
SEQUENCE DIAGRAM



A posture change drops the score below forty. The Policy Administrator kills the session at both enforcement points and revokes the outstanding certificate.

Policy Engine decision logic

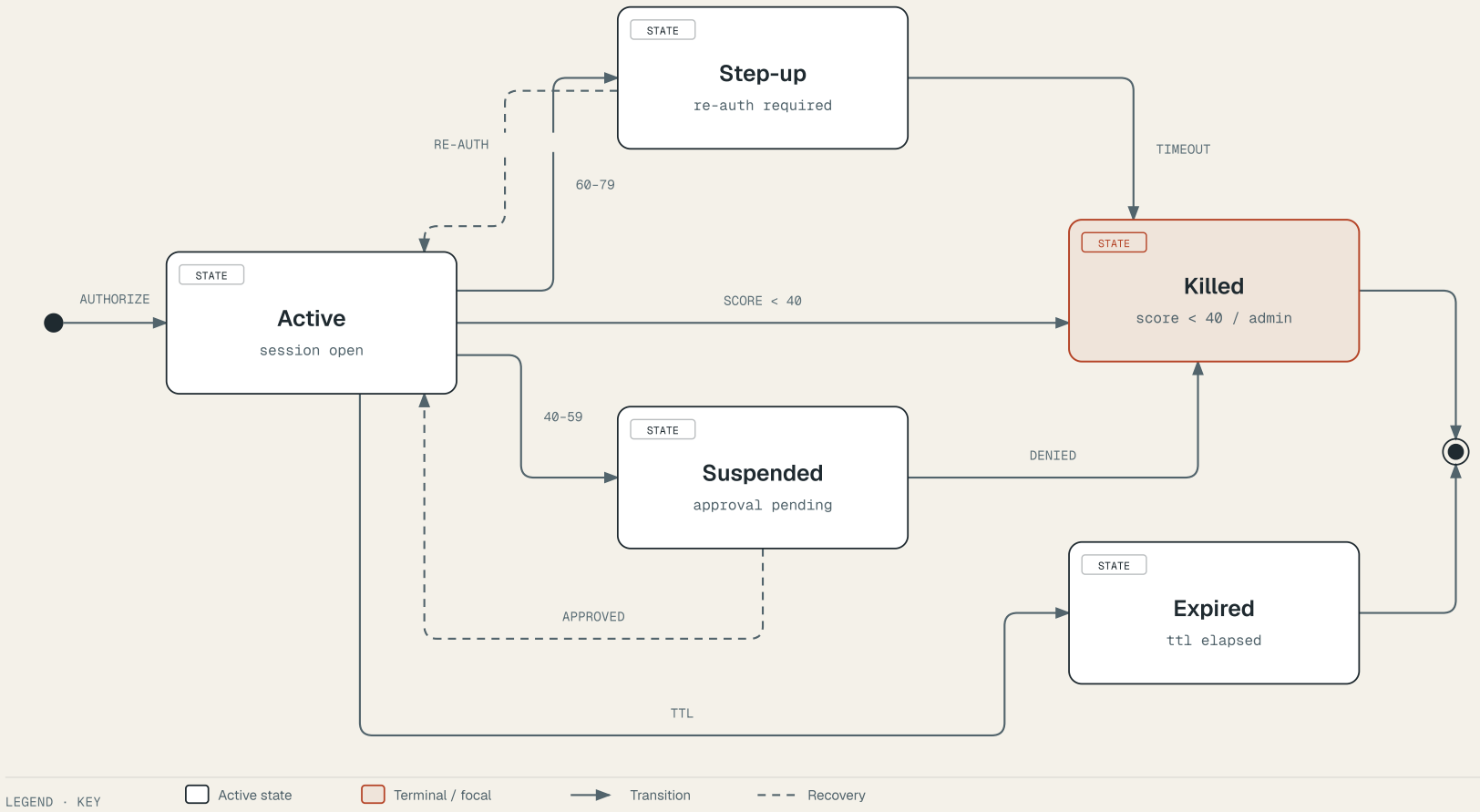
ACTIVITY DIAGRAM



Hard gates first, then rules plus the ML adjustment. The score selects one of four bands; a failed hard gate overrides the score and fails closed.

Session lifecycle

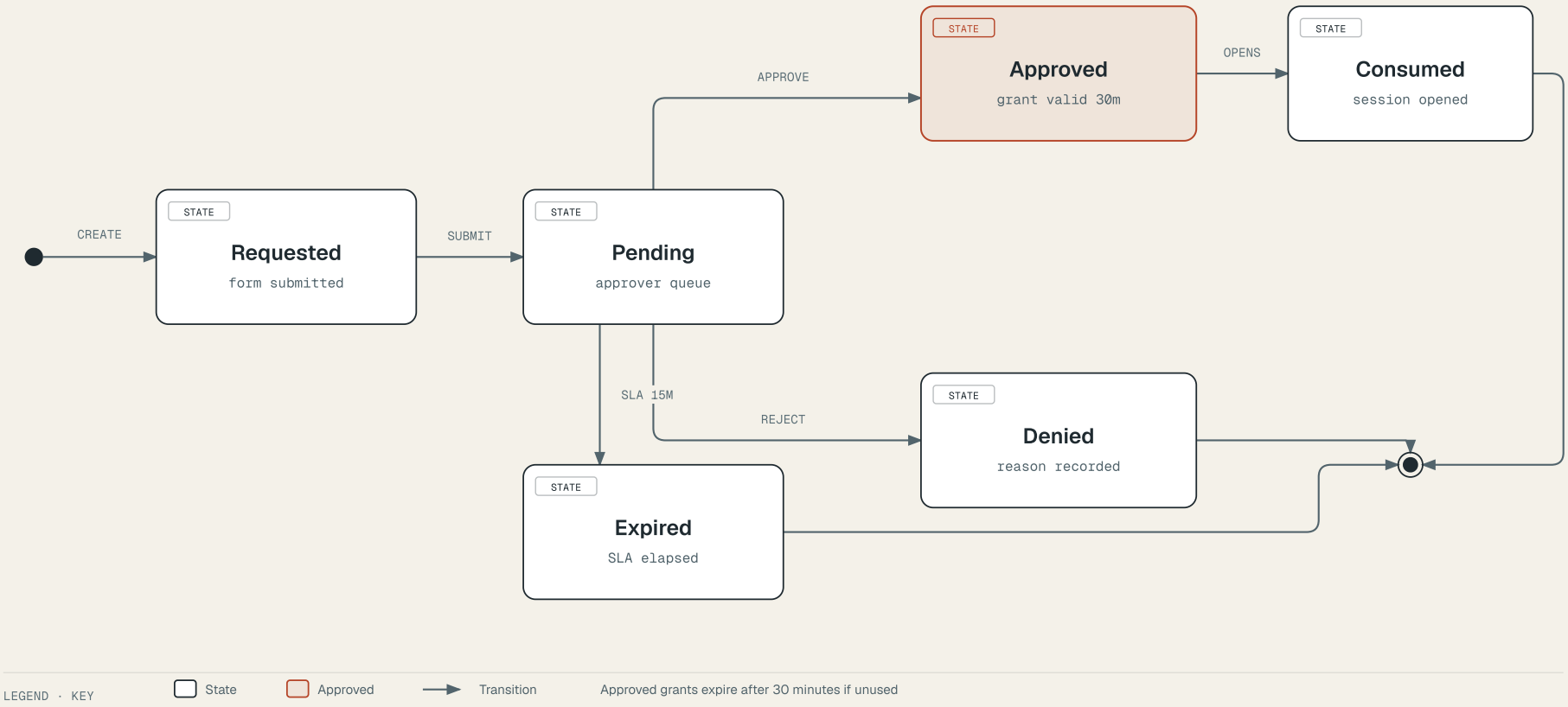
STATE MACHINE



Active, step-up, suspended, killed, and expired. Suspended waits for approval, step-up requires re-authentication, and killed and expired are terminal.

JIT request lifecycle

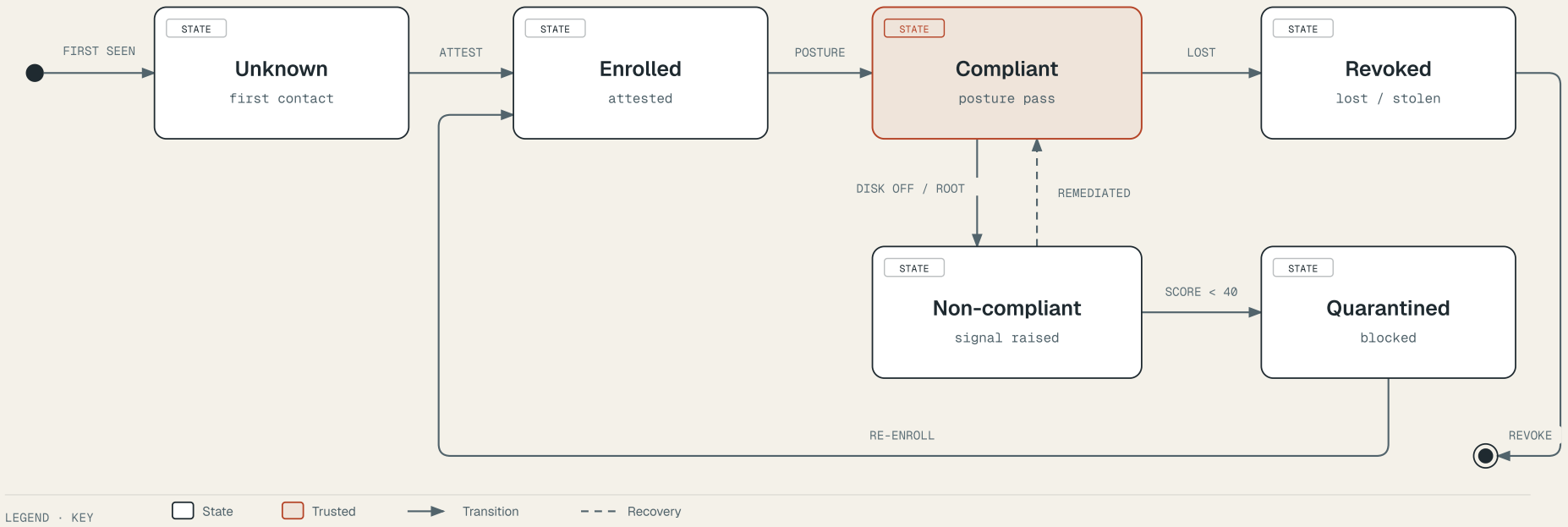
STATE MACHINE



Requested, pending, approved, denied, expired, and consumed. An approval is single-use and expires after thirty minutes if it is never consumed.

Device trust lifecycle

STATE MACHINE



Unknown, enrolled, compliant, non-compliant, quarantined, and revoked. A quarantined device re-enrolls only after the posture finding is remediated.